

STRATEGIC MIGRATION FRAMEWORK

PROJECT KOPERASI DIGITAL 2.0

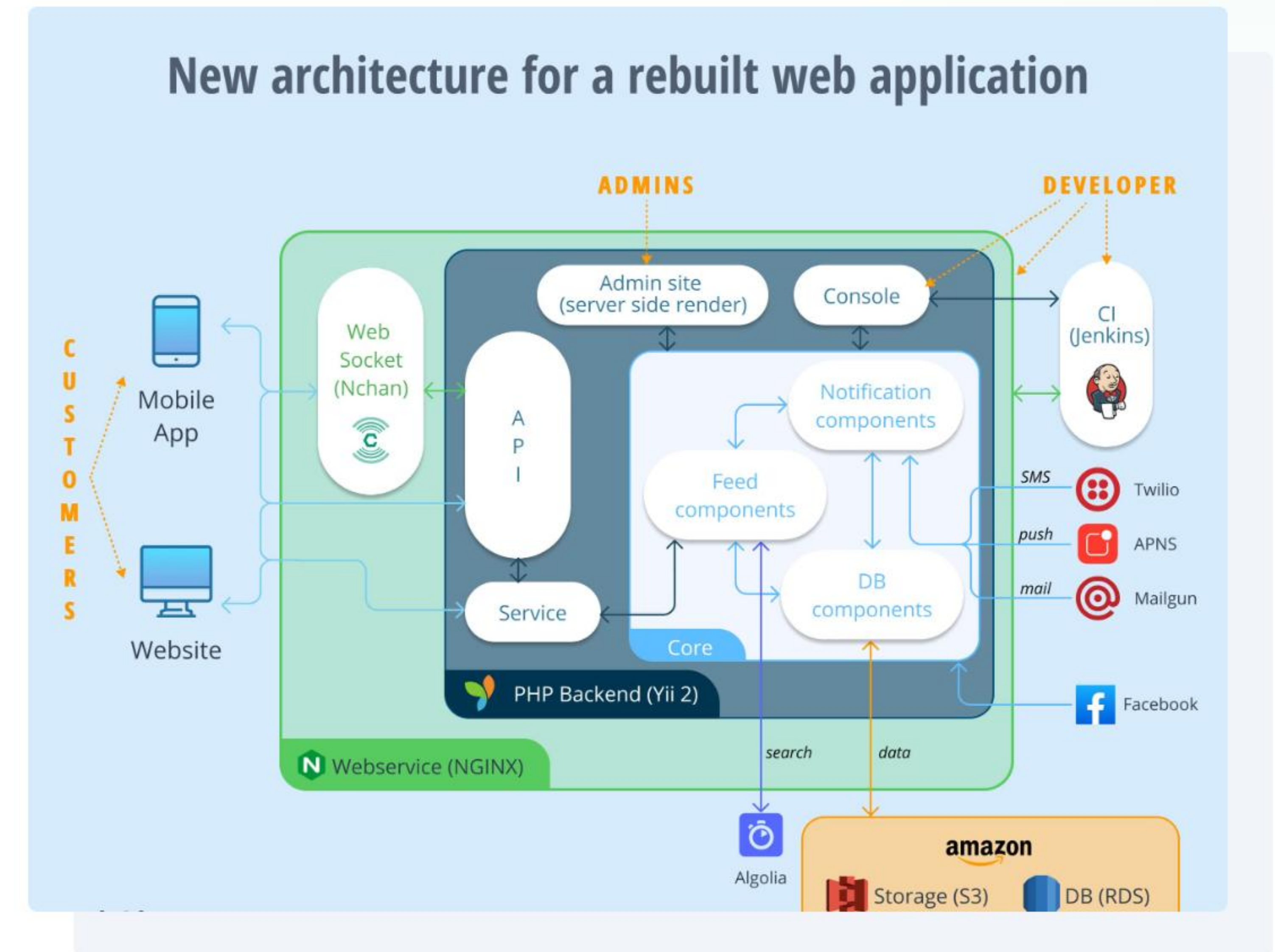
Full-Scale Migration to Next.js 15 Ecosystem: Modernizing Financial Governance and Member Liquidity.



THE TRANSITION STRATEGY

Shifting from **Laravel 1 Monolith** to a high-performance **Next.js 15 App Router** architecture.

- ⚡ **Performance:** Server Components (RSC) for instantaneous page loads.
- 🛡️ **Integrity:** Zero-compromise logic migration for financial ACID transactions.
- 📦 **Scalability:** Modular Feature-Sliced Design (FSD) for long-term maintenance.



THE MODERN FINANCIAL STACK



FRAMEWORK

Next.js 15 (App Router) with React 19 support for streaming and Server Actions.



PERSISTENCE

Prisma/Drizzle ORM mapping the 24 existing MySQL tables with strict TypeScript safety.



INTERFACE

Tailwind CSS paired with shadcn/ui for accessible, consistent, and dark-mode ready UI.

GUARDRAILS FOR INTEGRITY

- ✓ **ACID Transactions:** Financial calculations translated from PHP Services to precise TS logic in `lib/services/`.
 - 🗄️ **Schema Preservation:** Strict prohibition on modifying the existing 24-table core MySQL structure.
 - 👤 **RBAC Governance:** Auth.js middleware protecting routes for Member, Admin, and Superadmin roles.
 - 🔗 **Atomic Build Rules:** Every commit must pass `npm run build`, ensuring no breaking changes in production.
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ARCHITECTURE

SCALABLE RESTRUCTURE

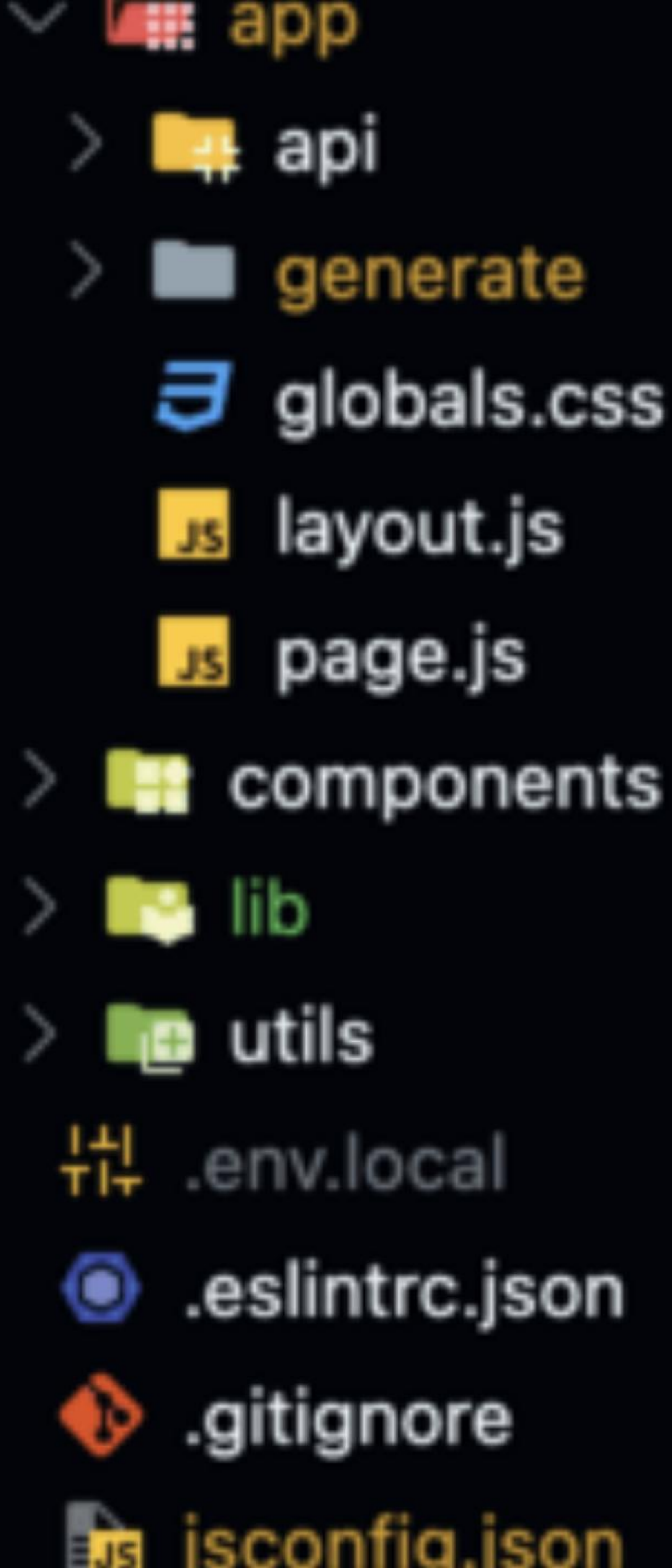
Moving from MVC to a Server-Action driven architecture. Clean separation between routes, logic, and data.

app/: Routing & Route Groups

lib/actions/: Mutative Server Actions

lib/services/: Pure Financial Logic

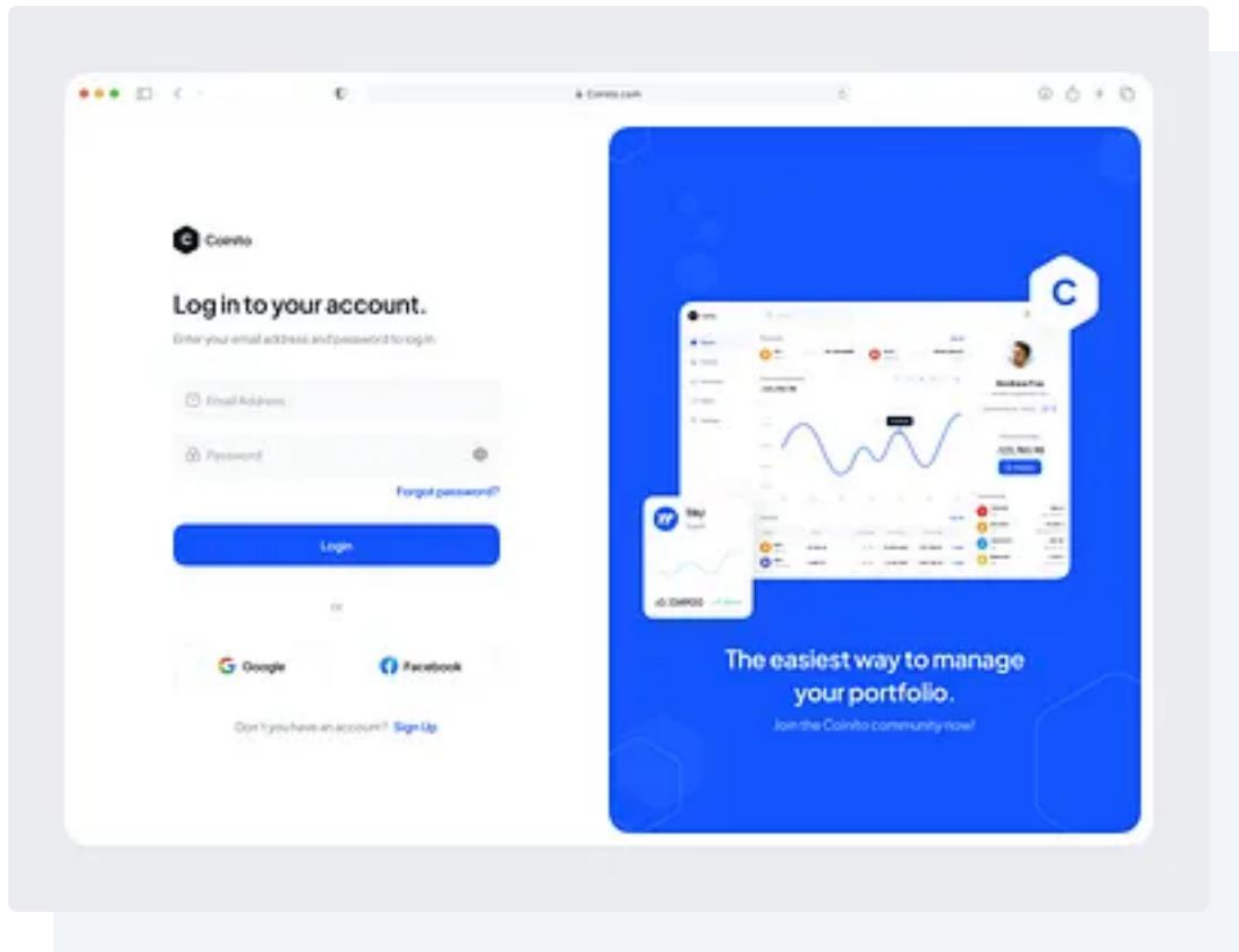
components/ui/: Standardized shadcn atoms



THE SOURCE OF TRUTH

Domain	Critical Tables (MySQL)	Migration Status
Core & Auth	units, members, users, sessions, audit_logs	Active
Savings	saving_types, savings, saving_transactions	Priority
Loans	loan_products, loan_applications, loans, schedules	Priority
Accounting	chart_of_accounts, journal_entries, journal_lines	Pending
Commerce	products, orders, stock_movements, ppob	Pending

USER-CENTRIC EXPERIENCE



MEMBER DASHBOARD

Real-time visibility into financial health, debt ratios, and SHU distributions.

Self-Service Loans: Direct application for Regular, Flash, and Goods-based loans.

Live SHU Tracking: Instant calculation of profit-sharing based on participation.

PPOB Hub: Utility bill payments (PLN, Data) directly from member balance.

FINANCIAL PRECISION

100%

ACID COMPLIANCE

0

SCHEMA BREAKAGE

Preserving mission-critical financial logic: Every interest calculation (Flat, Effective, Annuity) is mapped directly from legacy services to type-safe TypeScript functions.

COMMERCE & PPOB ECOSYSTEM

 Digital Payment Ecosystem Collage

DIGITAL ASSETS

Managing Token Listrik and Paket Data inventories as virtual product categories.



COOP SHOP (POS)

Kasbon integrated payments using the established "Paylater" ENUM for credit transactions.

ADMIN GOVERNANCE

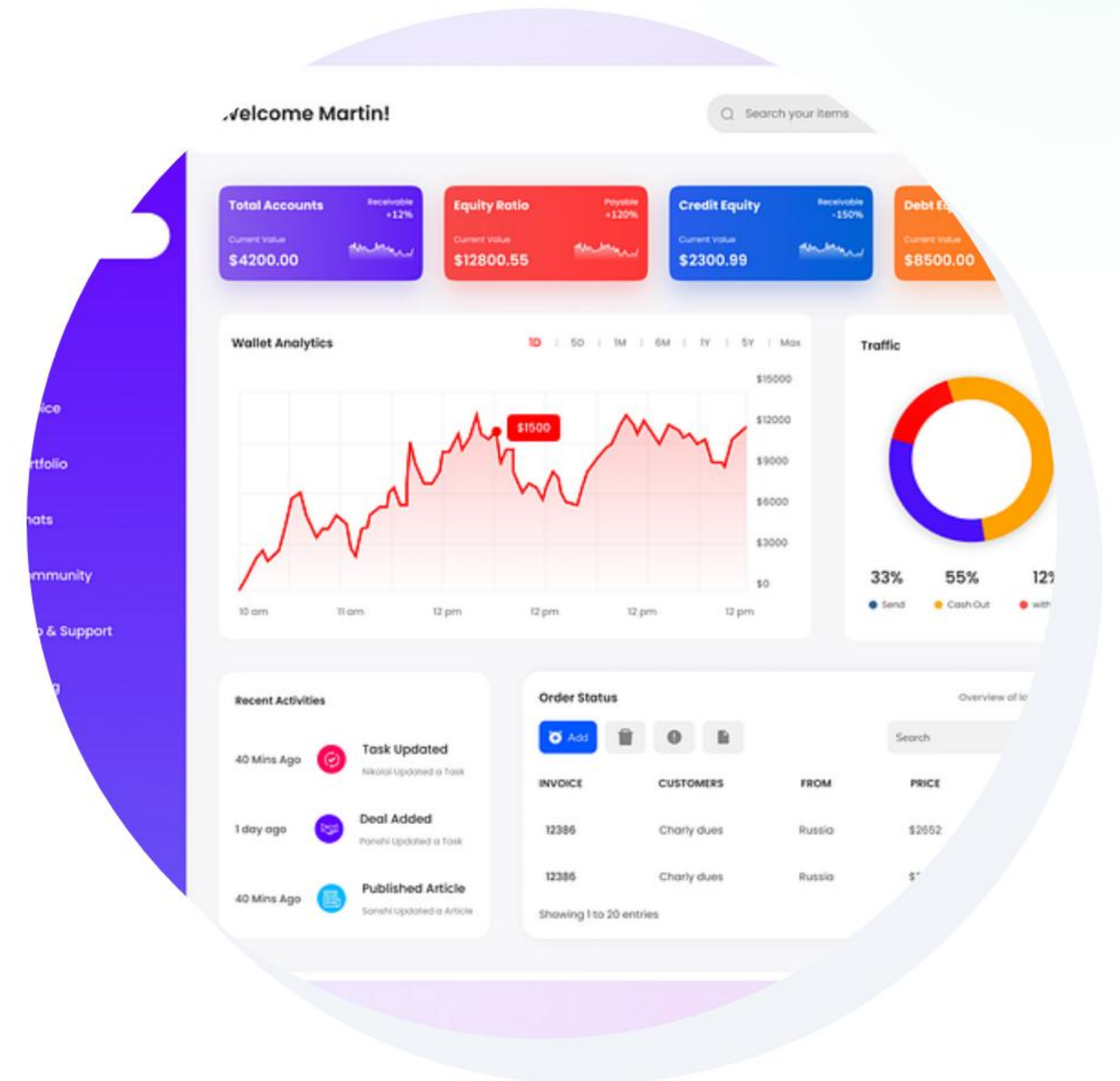
COMMAND CENTER

Comprehensive tools for management oversight and financial auditability.

Loan Approval: Multi-stage workflow for credit risk assessment.

Liquidity Management: Real-time balancing of Savings vs. Loan portfolios.

Audit Logs: Immutable tracking of every administrative action and configuration change.



DELIVERY ROADMAP



PHASE 1

Auth & Core
(Savings/Loans)



PHASE 2

Accounting
& SHU logic



PHASE 3

POS, PPOB
& Inventory



PHASE 4

Mobile Integration
& Deployment

QUESTIONS?

Koperasi Digital: The Next-Gen Financial Modernization.

Migration Authority | Full Next.js 15 Implementation

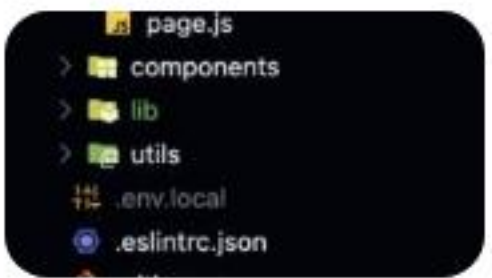


IMAGE SOURCES



<https://d2d1s3b58ci0ra.cloudfront.net/2024/07/new-architecture-after-legacy-application-migration.jpg>

Source: mobidev.biz



<https://preview.redd.it/best-practice-for-folder-structure-in-app-router-v0-5upaa9k2fl5d1.png?width=241&format=png&auto=webp&s=16aaaac2bc22917589dc78427b3d6554f38929eb>

Source: www.reddit.com



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Source: dribbble.com



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